

Firmware

Similar to the BIOS on your motherboard, all devices have their respective BIOSes, often called "the firmware". Some drives have upgradeable firmware, while others don't offer this feature for end users.

transfer, but if the file were around 20 GB, the difference would be considerable.

In the sequential write test, the Maxtor, once again, finished its job ahead of the rest, at 46.9 seconds. The IOmega was sluggish in this test, with 64.07 seconds. In the assorted write test, the IOmega HDD 160GB edged passed the rest, logging the lowest time—61.23 seconds. The Maxtor One Touch here was the slowest—it completed the task in 75.62 seconds.

Overall, it is the Maxtor One Touch that will be sufficient for the needs of users across the board, because in general, there will be both sequential and assorted data to be backed up and restored.

Conclusion

In this category the Western Digital Media Center topped the chart, being the only drive packed with features: it offered card readers, and USB and FireWire hub support.

It offered decent performance and a low price per MB of just 8 paise, which won it the Digit Best Buy Gold award.

The Maxtor, on the other hand, with its good features, better performance and solid build quality won the Silver Best Buy Award. It offered the lowest price per MB—just 6 paise—which makes it the most affordable backup solution.

The others were not far behind in either performance or features. Freecom would have topped the chart on the basis of its performance and build quality, but its price index of 11 paise per MB held it back.

Thin and Slim

There were a total of nine slim drives in this category. A few were quite good-looking, with capacities from 20 GB to an astonishing 80 GB. One wouldn't mind keeping them around on one's desktop. The Transcend TS0GHDC2 was the only odd-

capacity drive at 30 GB. They were all palm-sized, and a few, such as the Transcend TS20GSJ18 20GB, made cell phones look fat.

Features

The drives in question were eminently pocketable. The Transcend TS20GSJ18 20 GB and the TS40GSJ18 40 GB were the lightest drives, weighing just 140 grams each. Both these sported the body colour and make of the Apple iPod. The Freecom FHD-XS 20 GB and 40 GB models were the smallest external hard drives in the comparison. Both these were just 12 mm thick, and occupied the least space.

All nine drives sported USB 2.0, except for the Transcend TS0GHDC2 30 GB, which also had FireWire support. Of the nine drives, just three—all from Freecom—came with an application launch button. None of the Transcend drives had this.

Like the drives, the packages, too, were compact, and came with a quick-install guide, a software CD with an electronic manual, and an interface cable. That was all that was bundled with these drives. Only the Transcend TS0GHDC2 30GB came with a 6-pin to 4-pin FireWire cable converter.

All the drives drew power from the interface, which means that some older systems, especially laptops, might not be able to power them. To overcome this, all the Transcend 2.5-inch drives came with an extra power inlet. One can purchase a



The Freecom FHD-XS 40GB

power adapter separately if the need arises.

Build Quality

All the drives were sturdily built. The Transcend TS20GSJ18 felt the sturdiest. The Freecom FHD-XS came with a short, fixed USB cord, which is adequate for laptops—but a desktop PC might demand a USB extender cable.

Performance

In SiSoft Sandra 2004 Pro, the Transcend TS0GHDC2 30 GB drive gave the best drive index in this category—18,939. This, with a sequential read score of 28 MBps is comparable to any laptop drive. The random read score of 4 to 5 MB was scored by drives across the board, which is not bad, though the random read score for desktop drives is around 8 to 10 MB.

Both the Freecom FHD-XS drives returned poor results in SiSoft Sandra. They scored just half the score of the other drives, despite having the same USB 2.0 interface and 4,200 rpm drives. In access time, all the drives took 10 to 12 ms. This is



External Hard Drives

Brand	Model	Basic Specifications—Write speeds	Score	Price (Rs)
Desktop Hard Drives				
Freecom	FHD-1 120GB	120 GB, 3.5-inch platter, 900gm, 2MB buffer, USB 2.0/FireWire	72.78	14,000
IOmega	HDD 160GB	160 GB, 3.5-inch platter, 817gm, 8MB buffer, USB 2.0	72.19	10,950
Maxtor	OneTouch 250GB	250GB, 3.5-inch platter, 1400gm, 8MB buffer, USB 2.0/FireWire	73.37	16,500
Seagate	ST3160024A 160GB	160 GB, 3.5-inch platter, 900gm, 8MB buffer, USB 2.0/FireWire	71.84	11,000
Western Digital	Media Center	250 GB, 3.5-inch platter, 1450gm, 8MB buffer, USB 2.0/FireWire	73.75	19,500
Portable Hard Drives				
Freecom	FHD-XS 20GB	20 GB, 1.8-inch platter, 150gm, 2MB buffer, USB 2.0	44.60	15,000
Freecom	FHD-XS 40GB	40 GB, 1.8-inch platter, 150gm, 2MB buffer, USB 2.0	48.88	21,000
Freecom	FHD-2 PRO 80GB	80 GB, 1.8-inch platter, 175gm, 2MB buffer, USB 2.0	52.07	21,000
Transcend	StoreJet TS0GHDC2 30GB	30 GB, 2.5-inch platter, 240gm, 2MB buffer, USB 2.0/FireWire	49.37	9,000
Transcend	StoreJet TS20GSJ18 20GB	20 GB, 1.8-inch platter, 140gm, 2MB buffer, USB 2.0	40.36	9,000
Transcend	StoreJet TS40GSJ18 40GB	40 GB, 1.8-inch platter, 140gm, 2MB buffer, USB 2.0	47.83	11,000
Transcend	StoreJet TS0GSJ25 20GB	20 GB, 2.5-inch platter, 180gm, 2MB buffer, USB 2.0	50.37	7,800
Transcend	StoreJet TS0GSJ25 40GB	40 GB, 2.5-inch platter, 180gm, 2MB buffer, USB 2.0	47.92	14,330
Transcend	StoreJet TS0GSJ25 60GB	60 GB, 2.5-inch platter, 180gm, 2MB buffer, USB 2.0	50.67	16,900